# Principal Components Analysis ( PCA)

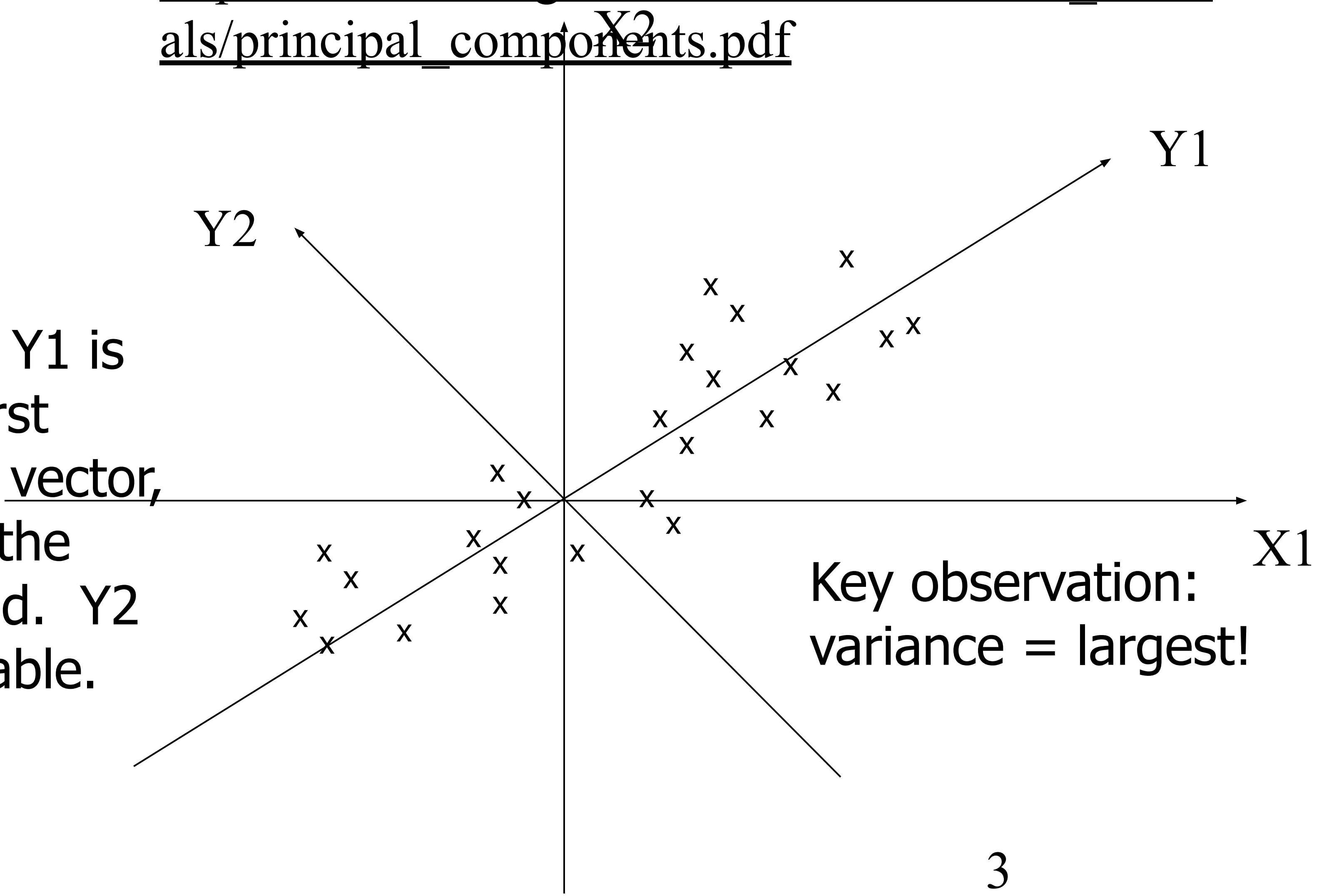
- An exploratory technique used to reduce the dimensionality of the data set to 2D or 3D
- Can be used to:
  - Reduce number of dimensions in data
  - Find patterns in high-dimensional data
  - Visualize data of high dimensionality
- Example applications:
  - Face recognition
  - Image compression
  - Gene expression analysis

# Principal Components Analysis Ideas ( PCA)

- Does the data set ‘span’ the whole of d dimensional space?
- For a matrix of *m* samples x *n* genes, create a new covariance matrix of size *n* x *n*.
- Transform some large number of variables into a smaller number of uncorrelated variables called principal components (PCs).
- developed to capture as much of the variation in data as possible

# Principal Component Analysis

- See online tutorials such as http://www.cs.otago.ac.nz/cosc453/student_tutorials/principal_components.pdf




Note: Y1 is the first eigen vector, Y2 is the second. Y2 ignorable.

Key observation: variance = largest!

# Principal Component Analysis: one attribute first

- Question: how much spread is in the data along the axis? (distance to the mean)
- Variance=Standard deviation^2

$$s^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}$$

| Temperature |
| ---: |
| 42 |
| 40 |
| 24 |
| 30 |
| 15 |
| 18 |
| 15 |
| 30 |
| 15 |
| 30 |
| 35 |
| 30 |
| 40 |
| 30 |

# Now consider two dimensions

Covariance: measures the correlation between X and Y

- cov(X,Y)=0: independent
- Cov(X,Y)>0: move same dir
- Cov(X,Y)<0: move oppo dir

$$\operatorname{cov}(X,Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{(n-1)}$$

| X=Temperature | Y=Humidity |
|---:|---:|
| 40 | 90 |
| 40 | 90 |
| 40 | 90 |
| 30 | 90 |
| 15 | 70 |
| 15 | 70 |
| 15 | 70 |
| 30 | 90 |
| 15 | 70 |
| 30 | 70 |
| 30 | 70 |
| 30 | 90 |
| 40 | 70 |
| 30 | 90 |

# More than two attributes: covariance matrix

- Contains covariance values between all possible dimensions (=attributes):

$$C^{nxn} = (c_{ij} \mid c_{ij} = \text{cov}(Dim_i, Dim_j))$$

- Example for three attributes (x,y,z):

$$C = \begin{pmatrix} \text{cov}(x,x) & \text{cov}(x,y) & \text{cov}(x,z) \\ \text{cov}(y,x) & \text{cov}(y,y) & \text{cov}(y,z) \\ \text{cov}(z,x) & \text{cov}(z,y) & \text{cov}(z,z) \end{pmatrix}$$

# Eigenvalues & eigenvectors

- Vectors **x** having same direction as $A$**x** are called *eigenvectors* of $A$ ($A$ is an n by n matrix).
- In the equation $A\mathbf{x}=\lambda\mathbf{x}$, $\lambda$ is called an *eigenvalue* of $A$.

$$\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} x \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 8 \end{pmatrix} = 4x \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

# Eigenvalues & eigenvectors

- $A\mathbf{x}=\lambda\mathbf{x} \Leftrightarrow (A\text{-}\lambda \mathrm{I})\mathbf{x}=0$
- How to calculate **x** and λ:
  - Calculate $det(A\text{-}\lambda I)$, yields a polynomial (degree n)
  - Determine roots to $det(A\text{-}\lambda I)=0$, roots are eigenvalues λ
  - Solve $(A\text{-}\lambda I)\,\mathbf{x}=0$ for each λ to obtain eigenvectors **x**

# Principal components

- 1. principal component (PC1)
  - The eigenvalue with the largest absolute value will indicate that the data have the largest variance along its eigenvector, the direction along which there is greatest variation
- 2. principal component (PC2)
  - the direction with maximum variation left in data, orthogonal to the 1. PC
- In general, only few directions manage to capture most of the variability in the data.

# Steps of PCA

- Let $\overline{X}$ be the mean vector (taking the mean of all rows)
- Adjust the original data by the mean
  $X' = X - \overline{X}$
- Compute the covariance matrix C of adjusted X
- Find the eigenvectors and eigenvalues of C.

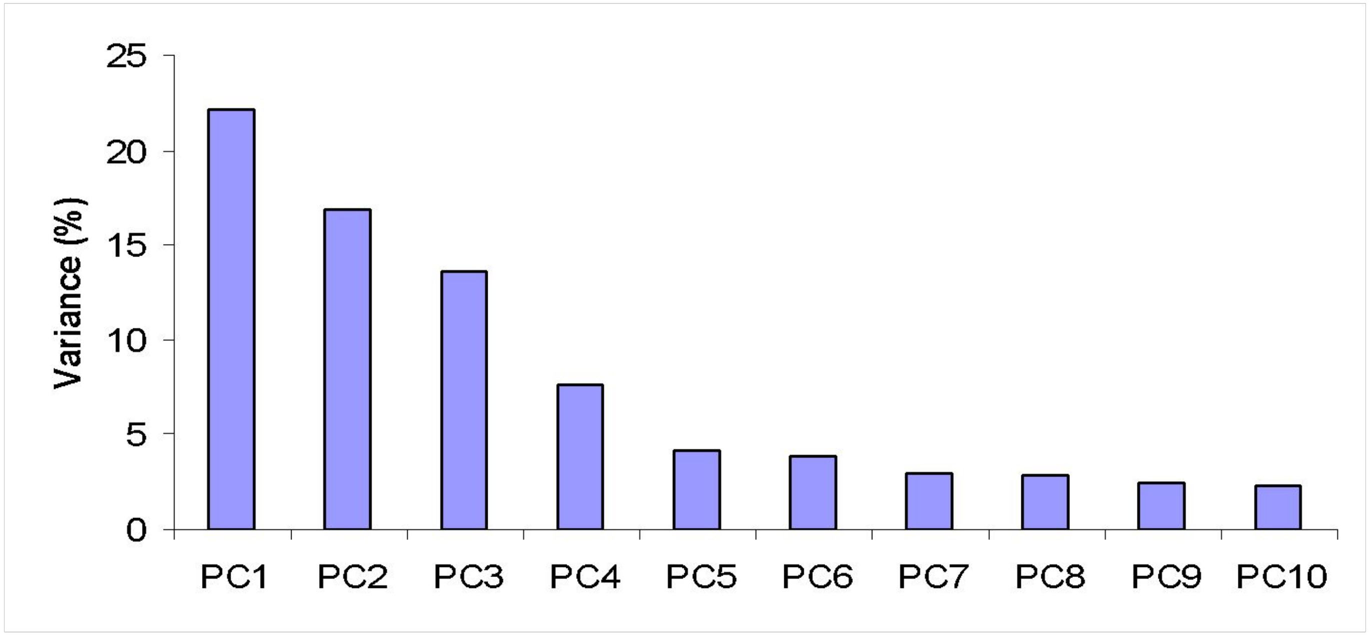
- For matrix $C$, vectors **e** (=column vector) having same direction as $C$**e** :
  - *eigenvectors* of $C$ is **e** such that $C\mathbf{e}=\lambda\mathbf{e}$,
  - $\lambda$ is called an *eigenvalue* of $C$.
- $C\mathbf{e}=\lambda\mathbf{e} \Leftrightarrow (C-\lambda I)\mathbf{e}=0$
  - **Most data mining packages do this for you.**

# Eigenvalues

- Calculate eigenvalues $\lambda$ and eigenvectors $\mathbf{x}$ for covariance matrix:
  - Eigenvalues $\lambda_j$ are used for calculation of [% of total variance] ($V_j$) for each component $j$:

$$V_j = 100 \cdot \frac{\lambda_j}{\sum_{x=1}^{n} \lambda_x} \qquad \sum_{x=1}^{n} \lambda_x = n$$

# Principal components - Variance



# Transformed Data

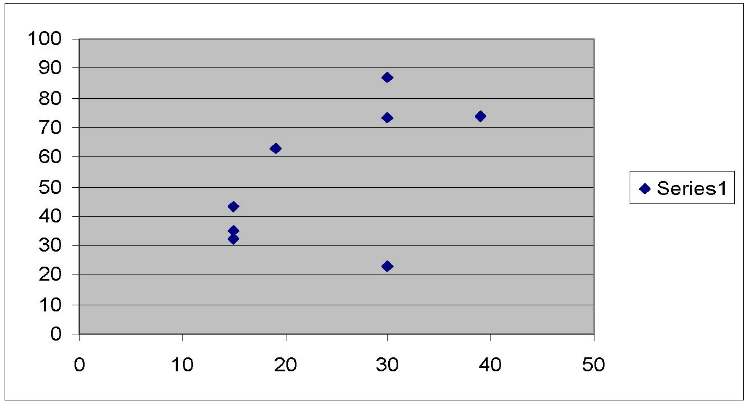
- Eigenvalues $\lambda_j$ corresponds to variance on each component $j$
- *Thus, sort by* $\lambda_j$
- Take the first $p$ eigenvectors $\mathbf{e}_{\mathbf{i};}$ where p is the number of top eigenvalues
- These are the directions with the largest variances

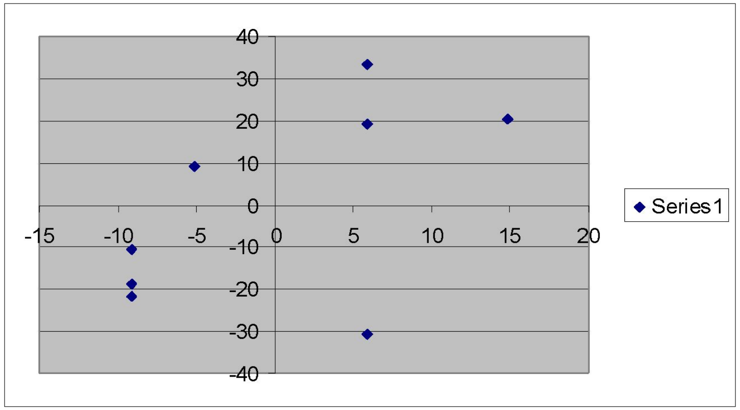
$$\begin{pmatrix} y_{i1} \\ y_{i2} \\ \dots \\ y_{ip} \end{pmatrix} = \begin{pmatrix} e_1 \\ e_2 \\ \dots \\ e_p \end{pmatrix} \begin{pmatrix} x_{i1} - \overline{x_1} \\ x_{i2} - \overline{x_2} \\ \dots \\ x_{in} - \overline{x_n} \end{pmatrix}$$

# An Example

Mean1=24.1
Mean2=53.8

| X1 | X2 | X1' | X2' |
|---|---|---|---|
| 19 | 63 | -5.1 | 9.25 |
| 39 | 74 | 14.9 | 20.25 |
| 30 | 87 | 5.9 | 33.25 |
| 30 | 23 | 5.9 | -30.75 |
| 15 | 35 | -9.1 | -18.75 |
| 15 | 43 | -9.1 | -10.75 |
| 15 | 32 | -9.1 | -21.75 |
| 30 | 73 | 5.9 | 19.25 |



# Covariance Matrix

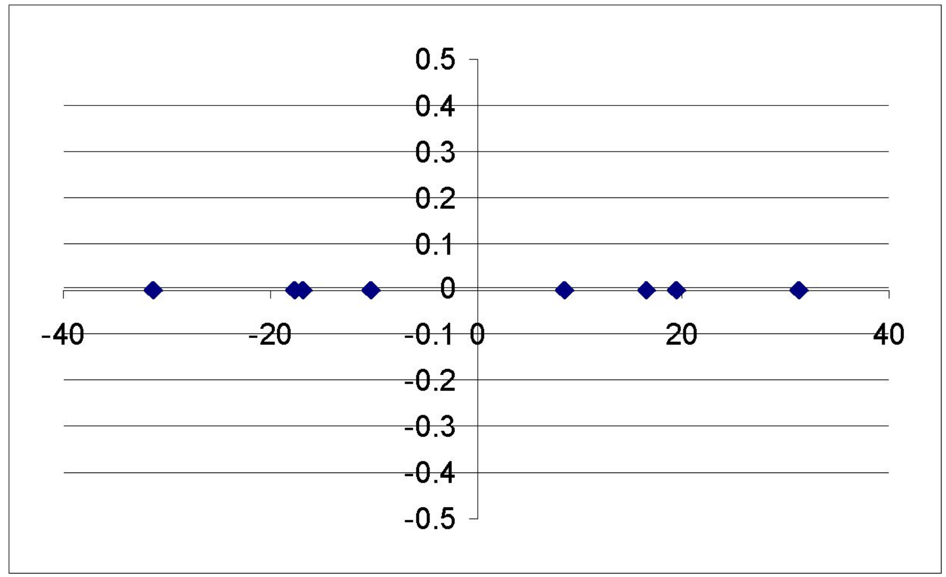
- C=

| | |
|---|---|
| 75 | 106 |
| 106 | 482 |

- Using MATLAB, we find out:
  - Eigenvectors:
  - e1=(-0.98,-0.21), $\lambda 1=51.8$
  - e2=(0.21,-0.98), $\lambda 2=560.2$
  - Thus the second eigenvector is more important!

# If we only keep one dimension: e2

- We keep the dimension of e2=(0.21,-0.98)
- We can obtain the final data as




| yi |
|---|
| -10.14 |
| -16.72 |
| -31.35 |
| 31.374 |
| 16.464 |
| 8.624 |
| 19.404 |
| -17.63 |

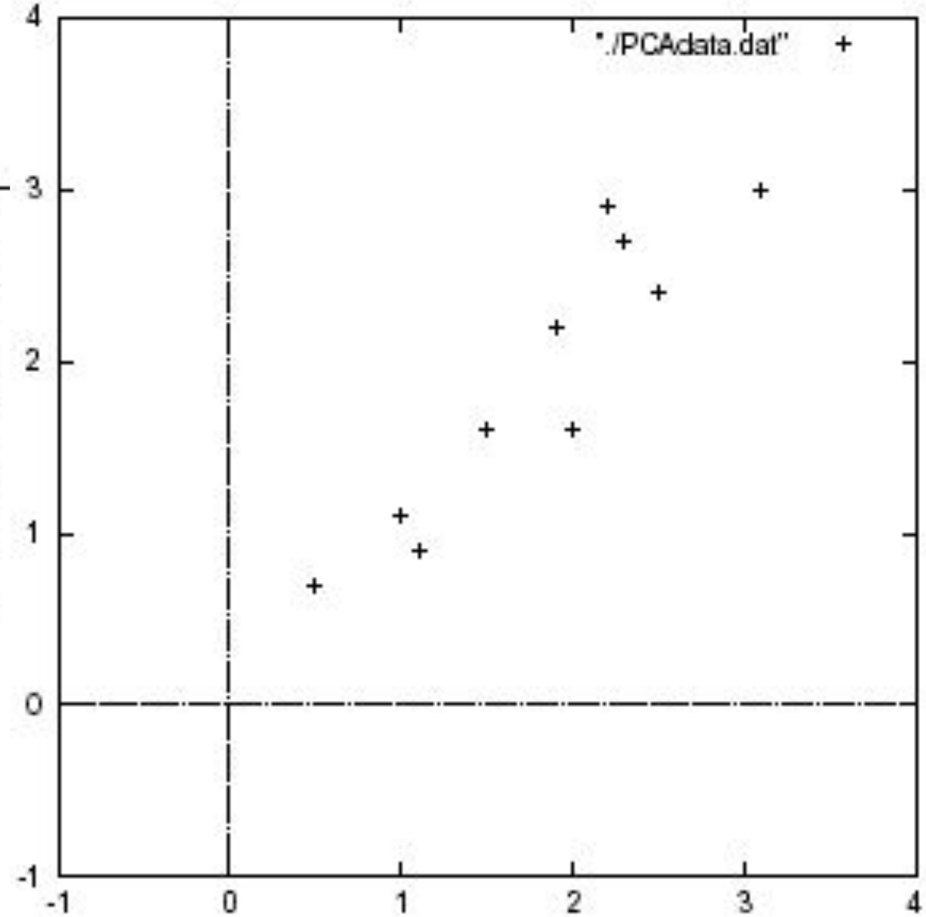
$$y_i = \begin{pmatrix} 0.21 & -0.98 \end{pmatrix} \begin{pmatrix} x_{i1} \\ x_{i2} \end{pmatrix} = 0.21 * x_{i1} - 0.98 * x_{i2}$$

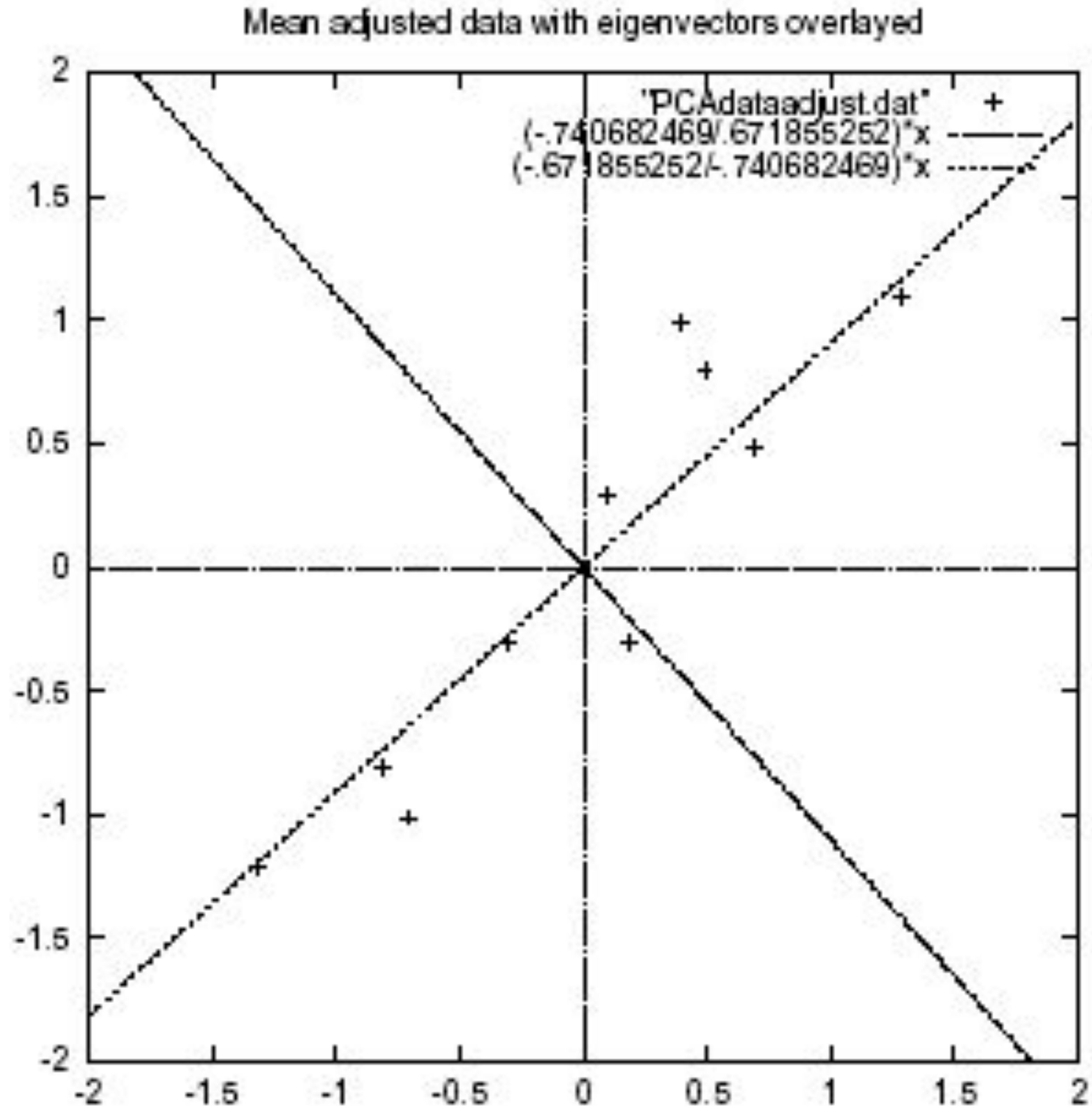
Data =

| $x$ | $y$ |
|---|---|
| 2.5 | 2.4 |
| 0.5 | 0.7 |
| 2.2 | 2.9 |
| 1.9 | 2.2 |
| 3.1 | 3.0 |
| 2.3 | 2.7 |
| 2 | 1.6 |
| 1 | 1.1 |
| 1.5 | 1.6 |
| 1.1 | 0.9 |

DataAdjust =

| $x$ | $y$ |
|---|---|
| .69 | .49 |
| -1.31 | -1.21 |
| .39 | .99 |
| .09 | .29 |
| 1.29 | 1.09 |
| .49 | .79 |
| .19 | -.31 |
| -.81 | -.81 |
| -.31 | -.31 |
| -.71 | -1.01 |

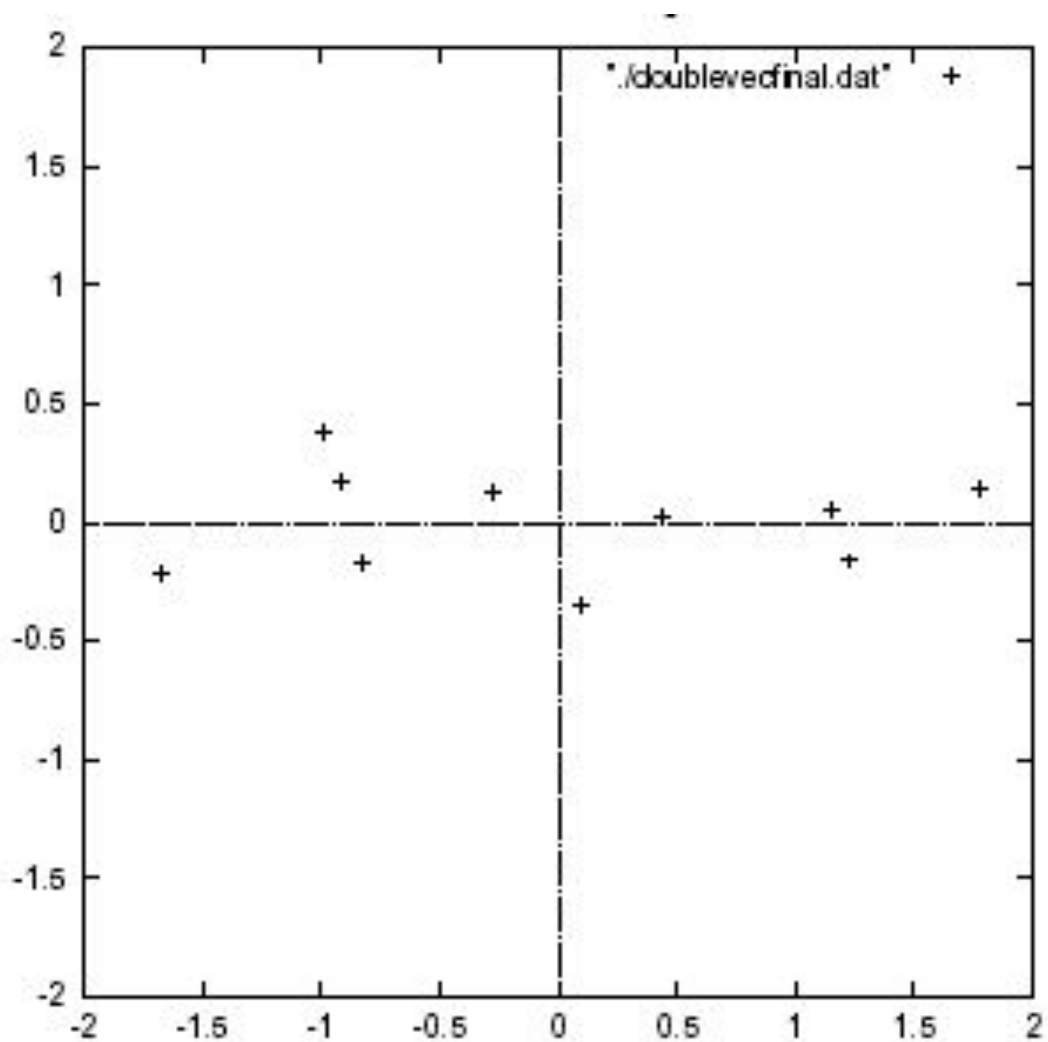



Mean adjusted data with eigenvectors overlayed
"PCAdataadjust.dat"
(-.740682469/.671855252)*x
(-.671855252/-.740682469)*x
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2

Transformed Data=

| $x$ | $y$ |
|---|---|
| -.827970186 | -.175115307 |
| 1.77758033 | .142857227 |
| -.992197494 | .384374989 |
| -.274210416 | .130417207 |
| -1.67580142 | -.209498461 |
| -.912949103 | .175282444 |
| .0991094375 | -.349824698 |
| 1.14457216 | .0464172582 |
| .438046137 | .0177646297 |
| 1.22382056 | -.162675287 |

Data transformed with 2 eigenvectors



# PCA –> Original Data

- Retrieving old data (e.g. in data compression)
  - $RetrievedRowData = (RowFeatureVector^{T} \times FinalData) + OriginalMean$
  - Yields original data using the chosen components

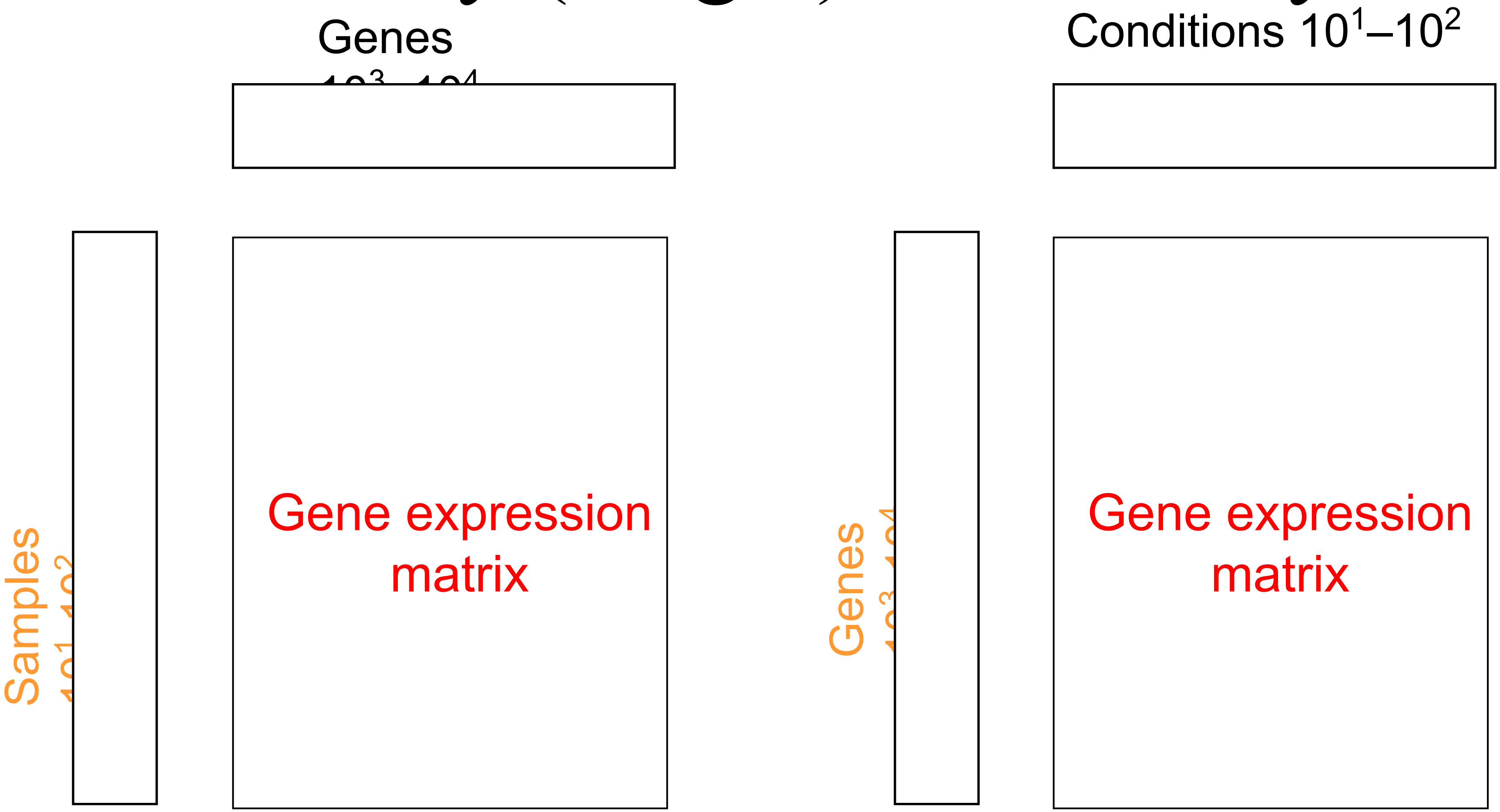
# Principal components

- General about principal components
  - summary variables
  - linear combinations of the original variables
  - uncorrelated with each other
  - capture as much of the original variance as possible

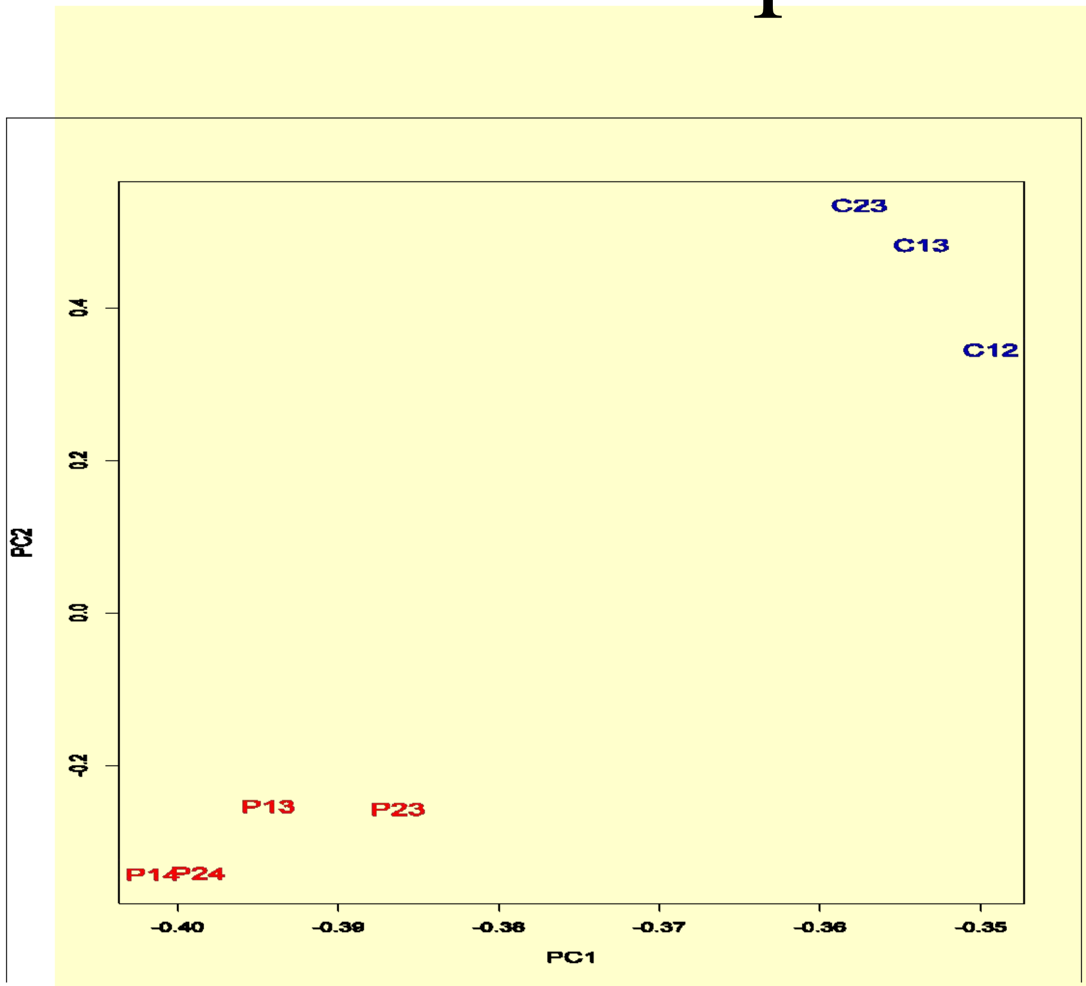
# Applications – Gene expression analysis

- Reference: Raychaudhuri et al. (2000)
- **Purpose:** Determine core set of conditions for useful gene comparison
- Dimensions: conditions, observations: genes
- Yeast sporulation dataset (7 conditions, 6118 genes)
- **Result:** Two components capture most of variability (90%)
- Issues: uneven data intervals, data dependencies
- PCA is common prior to clustering
- Crisp clustering questioned : genes may correlate with multiple clusters
- Alternative: determination of gene's closest neighbours

# Two Way (Angle) Data Analysis



# PCA - example

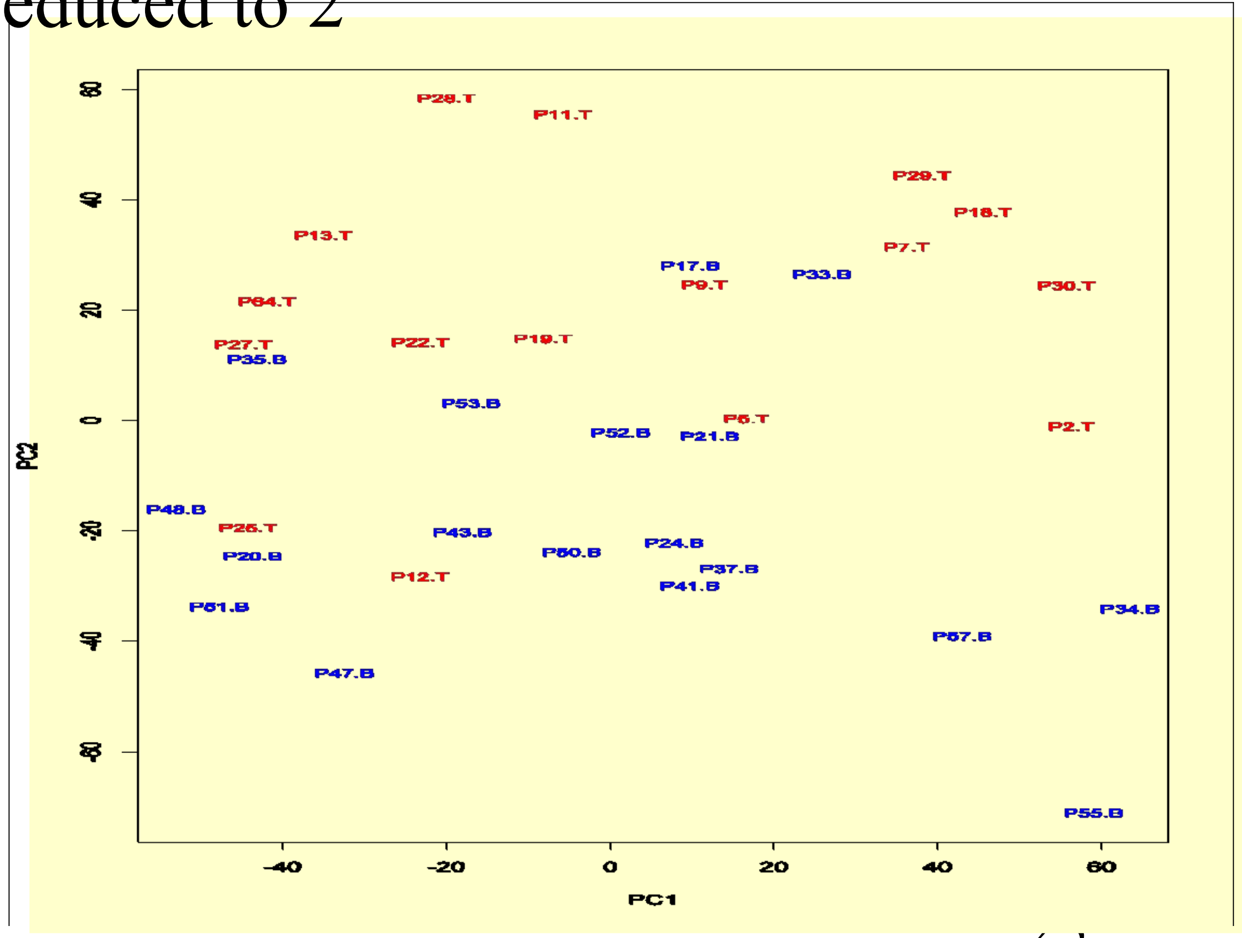


C23
C13
C12
P13
P23
P14
P24
0.4
0.2
0.0
-0.2
PC2
-0.40
-0.39
-0.38
-0.37
-0.36
-0.35
PC1

# PCA on all Genes

## Leukemia data, precursor B and T

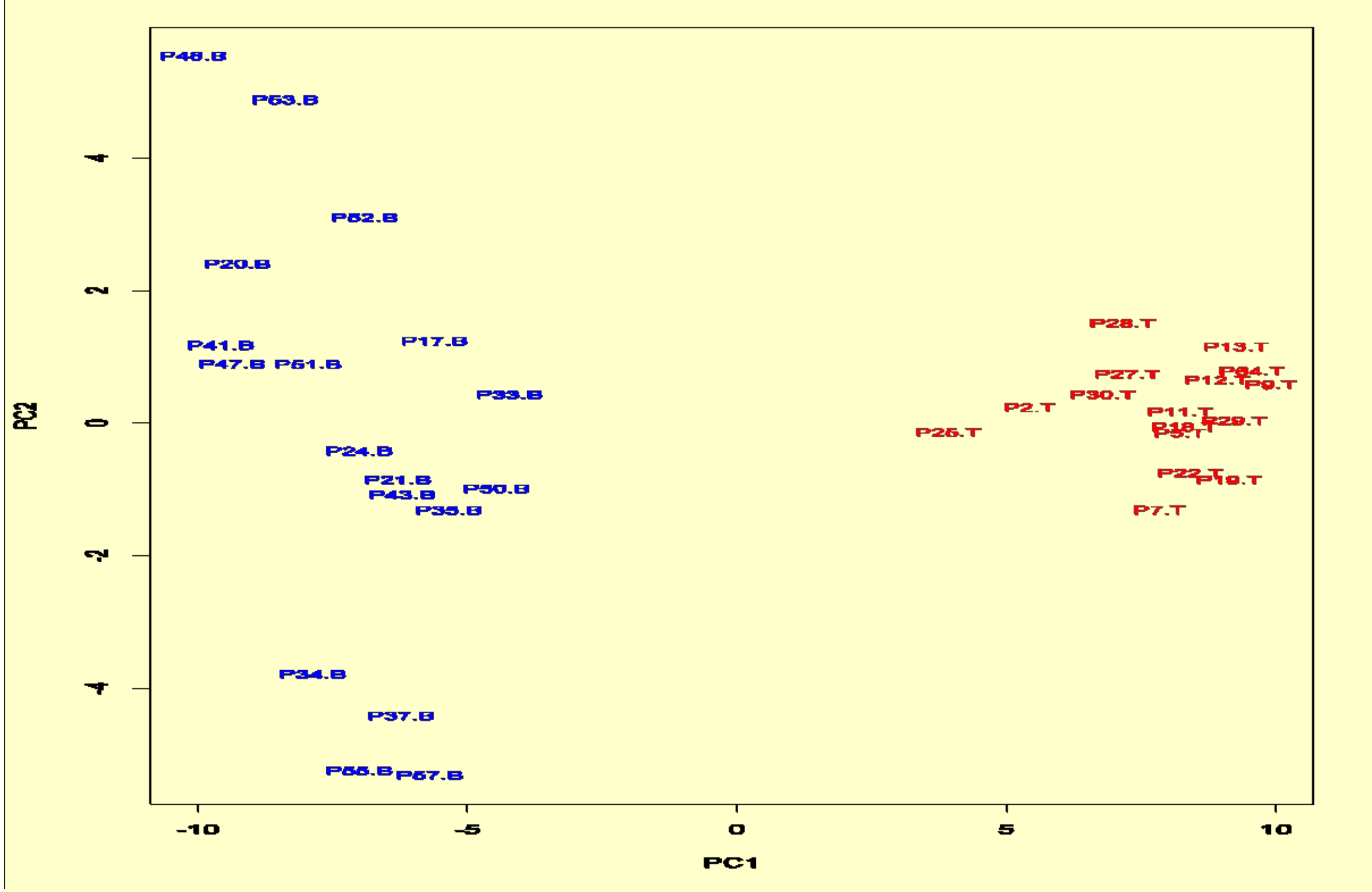
Plot of 34 patients, dimension of 8973 genes reduced to 2



# PCA on 100 top significant genes

## Leukemia data, precursor B and T

Plot of 34 patients, dimension of 100 genes reduced to 2



# PCA of genes (Leukemia data)

Plot of 8973 genes, dimension of 34 patients reduced to 2